

RESEARCH DESIGN

Step 1: Consider Aims and Data Requirement

What is the research question?

What data need to be collected to answer the research question?

Step 2: Consider aims and approach

What approach to adopt to collect the data?

Qualitative or Quantitative data

Practical and ethical considerations when designing research

, one needs to consider [research ethics](#).

- How much time is needed to collect data?
- Feasibility to gain access needed to get the data (e.g., by travelling to a specific location or contacting specific people)?
- Necessary research skills (e.g., [statistical analysis](#) or interview techniques)?
- Ethical approval, If research involves people or animals

At each stage of the research design process, make sure that your choices are practically feasible.

Types of quantitative research designs

Quantitative designs can be split into four main types.

Experimental

Experimental

- Used to test causal relationships
- Involves manipulating an independent variable and measuring its effect on a dependent variable
- Subjects are randomly assigned to groups
- Usually conducted in a controlled environment (e.g., a lab)

Observation

- Used to test whether (and how strongly) variables are related
- Variables are measured without influencing them

Descriptive

- Used to describe characteristics, averages, trends, etc
- Variables are measured without influencing them

Types of qualitative research designs

Qualitative designs are flexible and less strictly defined.

Case study

- Detailed study of a specific subject (e.g., a place, event, organization, etc).
- Data can be collected using a variety of sources and methods.
- Focuses on gaining a holistic understanding of the case.

Ethnography

- Detailed study of the culture of a specific community or group.
- Data is collected by extended immersion and close observation.
- Focuses on describing and interpreting beliefs, conventions, social dynamics, etc.

Grounded theory

- Aims to develop a theory inductively by systematically analyzing qualitative data.

Step 3: Identify Population and Sampling Method

A population is the entire group that for which researcher wants to draw conclusions about, while a sample is the smaller group representing the population.

Defining the population

For example,

- people from a specific demographic, region or background
- people with a certain job or medical condition
- users of a particular product
- Engg. Frequency range, voltage range

Sampling methods

Even with a narrowly defined population, it is difficult to get data on all entities.

There are two main approaches: probability sampling and non-probability sampling.

Probability sampling

- Sample is selected using random methods
- Mainly used in quantitative research
- Allows you to make strong statistical inferences about the population

Non-probability sampling

- Sample selected in a non-random way
- Used in both qualitative and quantitative research
- Easier to achieve, but chance of research bias

Case selection in qualitative research

In some types of qualitative designs, sampling may not be relevant.

For example, in a case study, the aim is to deeply understand a specific context, not to generalize to a population.

Careful selection of cases relevant to research question.

Step 4: Choose data collection methods

Data collection methods are ways of directly measuring variables and gathering information. They allow first-hand knowledge and insights into research problem.

Survey methods

Surveys allow you to collect data about opinions, behaviors, experiences, and characteristics by asking people directly. There are two main survey methods:

Questionnaires Interviews

Observation methods

Observational studies allow data collect real life situation without causing obstruction.

Observations may be conducted in real time, taking notes as you observe, or you might make audiovisual recordings for later analysis. They can be qualitative or quantitative.

Other methods of data collection

There are many other ways you might collect data depending on field and topic.

- Media & communication: Collecting a sample of texts (e.g., speeches, articles, or social media posts) for data on cultural norms and narratives
- Psychology: Using technologies like neuroimaging, eye-tracking, or computer-based tasks to collect data on things like attention, emotional response, or reaction time

- Education: Using tests or assignments to collect data on knowledge and skills
- Physical sciences: Using scientific instruments like weight, blood pressure, or chemical composition
- Video and image for data processing

Secondary data

Data that is already available at various sources.—for example, datasets from government surveys or previous studies on your topic.

Step 5: Plan Data Collection Procedures

Plan exactly how data collection methods be used to collect data in a consistent, accurate, and unbiased manner.

Operationalization

Survey

What questions to ask and what range of responses to be offered? -> Design of Questionnaire

Example

To measure teachers' satisfaction with online learning tools, you could create a questionnaire with a 5-point rating scale.

Observation

Some variables, like height or age, are easily measured. But often with more abstract concepts, like satisfaction, anxiety, or competence. Operationalization means turning these fuzzy ideas into measurable indicators.

Example

To measure student participation in an online course, you could record the number of times students ask and answer questions.

Experiment

-> Design of experimental setup

Reliability and validity

Reliability means your results can be consistently reproduced, while validity means that you're actually measuring the concept you're interested in.

Sampling procedures

How sampling to be done so the it is easy to carry out and is representative of the population

Data management

It's also important to create a data management plan for organizing and storing your data.

Step 6: Decide on Data Analysis Strategies

Raw Data Cannot answer the Research Question. Data Has To be analysed to draw inference.

Quantitative data analysis

In quantitative research, use some form of statistical analysis.

Descriptive statistics: Summarize sample data.

The distribution of the data (e.g., the frequency of each score on a test)

The central tendency of the data (e.g., the mean to describe the average score)

The variability of the data (e.g., the standard deviation to describe how spread out of scores)

The specific calculations depend on the level of measurement of your variables.

Inferential Statistics: Make estimates about the population based on sample data.

Test hypotheses about a relationship between variables.

Regression and correlation tests look for associations between two or more variables, while comparison tests (such as t tests and ANOVAs) look for differences in the outcomes of different groups.

Qualitative data analysis

Identify patterns, extract the parts that are most relevant and interpret its meanings to the research question.